Tobias Mwangi-Hartley

Data Scientist | Applied Machine Learning

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PROFILE

Data scientist with 6 years of experience in analytics and data science, including 4+ years as a data scientist. Builds and evaluates machine-learning models and scheduled batch workflows for forecasting, churn targeting and experimentation. Strong Python, statistical modelling and feature engineering, with experience using BigQuery, Vertex AI Workbench, dbt and automated tests. Moving towards machine learning engineering, with a focus on developing production software engineering practice.

TECHNICAL SKILLS

Python (pandas, scikit-learn, LightGBM, XGBoost, statsmodels, pytest), SQL (BigQuery), R (tidyverse, brms), dbt, Git, Jupyter, Looker, Vertex AI Workbench, Docker, Streamlit, PyTorch (course project)
Machine learning: Feature engineering, time-series forecasting, churn propensity modelling, model evaluation, SHAP, Bayesian statistics
Experimentation: A/B testing, sequential testing, causal inference

EXPERIENCE

Data Scientist | Thistlebox (meal-kit subscriptions), Edinburgh
Feb 2022–Present

- Built a weekly, six-week-ahead recipe and region demand forecasting model using LightGBM and calendar, price and menu-position features. Its WAPE improved from 23% to 14%; food waste decreased by an estimated 9%.
- Operationalised forecasts through a scheduled Vertex AI Workbench notebook writing results to BigQuery for operations use in Looker.
- Developed an XGBoost churn propensity model, scored weekly in BigQuery, to target retention offers to top-decile customers. A 10% holdout validated a 2.1 percentage-point reduction in churn in the treated group.
- Design and analyse approximately 40 A/B tests annually for growth and product teams; built a sequential testing calculator used by around 25 colleagues.
- Wrote the Streamlit experiment-planning app and its Dockerfile; the data platform team handles deployment to Cloud Run.
- Since 2025, added pytest unit tests for forecasting feature code, reaching approximately 60% coverage.
- Write production SQL in BigQuery and maintain dbt models for the analytics layer; present results to leadership monthly.
- Delivered the internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail, Glasgow
Sep 2020–Jan 2022

- Performed customer segmentation and tariff analysis using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

PROJECTS

Image Classification Course Project | Personal, 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset, achieving approximately 80% accuracy as part of an online deep-learning course.
- Tools: PyTorch, torchvision, Google Colab.

EDUCATION

MSc Statistics, University of the Lothians | 2020 | Distinction
Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics, University of the Lothians | 2019

LANGUAGES

English (native) | Swahili (conversational)